

Week 4 Final Report: Security Hardening and Best Practices

System Under Audit: Metasploitable2

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Objective:

To strengthen the overall security of Metasploitable2 by addressing previously identified vulnerabilities through system hardening measures and validation through security auditing.

1. Review of Previous Reports (Weeks 1–3):

- **Week 1: Information Gathering**
 - Discovered open ports (21, 22, 23, 25, 53, 80, etc.)
 - Identified outdated and vulnerable services: vsftpd 2.3.4, OpenSSH 4.7p1, Apache 2.2.8
- **Week 2: Vulnerability Assessment**
 - Critical vulnerabilities found in FTP, Telnet, Samba, and PostgreSQL
 - Lack of firewall and intrusion detection
- **Week 3: Exploitation**
 - Successfully exploited:
 - vsftpd backdoor (port 21)
 - Telnet login with weak credentials
 - Samba user enumeration and file access
 - PostgreSQL access without authentication

2. Hardening Measures Applied:

Task	Command(s)
System update	<code>sudo apt-get update && sudo apt-get upgrade -y</code>
Disable Telnet	<code>systemctl stop telnet</code> <code>systemctl disable telnet</code>

Task	Command(s)
Remove vsftpd	apt-get remove vsftpd -y
Setup UFW Firewall	ufw allow OpenSSH ufw enable ufw status
Install Snort (IDS)	apt-get install snort -y
Setup AIDE (File Integrity)	apt-get install aide -y aideinit
Enforce Password Policy	Edited /etc/pam.d/common-password to enforce complexity and length



3. Post-Hardening Audit Results:

Tools Used:

- Nmap
- Manual config check
- Metasploit validation

Key Findings Post-Hardening:

Service	Status	Notes
FTP (vsftpd)	Removed	Port 21 closed
Telnet	Disabled	Port 23 inaccessible
Samba	Still running	To be hardened in next steps
PostgreSQL	Hardened	Enforced auth and restricted access
Firewall	Enabled	Blocking unused ports
IDS (Snort)	Active	Monitoring common attack vectors
AIDE	Initialized	Ready to detect unauthorized changes



4. Improvements Achieved:

- Closed critical open ports (FTP, Telnet)
- Removed backdoor-prone services

- Improved access control and monitoring
- Ensured minimal attack surface
- Improved visibility using Snort and AIDE

 5. Remaining Risks & Recommendations:

Risk	Recommendation
Samba vulnerabilities	Consider removing or configuring securely
Weak application design	Avoid deploying real applications on vulnerable OS
No automatic patching	Schedule regular updates manually

 Conclusion:

The system has been significantly hardened by applying foundational security practices. Key vulnerabilities exploited earlier were mitigated, and essential monitoring and protection tools were introduced. Further steps can include continuous monitoring and penetration testing to maintain security posture.